

Pushpanath M R

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Experience

5G L1 (PHY Layer) Intern, Nokia Solutions & Networks – Bengaluru, India Aug 2025 – Jun 2026

- Designed enhancements to the SCT (System Component Test) framework with scalable test scenario handling and MATLAB-based generation, reducing test suite file sizes by **40%** and improving validation pipeline throughput.
- Collaborated with cross-functional RAN Integration teams in an **Agile** environment, gaining exposure to end-to-end PUCCH and network validation workflows beyond L1 PHY scope.

Full-Stack Developer Intern, CodeStreams Innovations Pvt Ltd – Bengaluru, India Aug 2023 – Oct 2023

- Engineered 5+ responsive web interfaces using React.js, Next.js, and Tailwind CSS, reducing UI inconsistencies by **60%** through a standardized component library.
- Configured Firebase Auth with real-time database synchronization (**100+ concurrent sessions**, 99.9% uptime) and validated 10+ REST API endpoints using Postman, resolving 4 critical edge-case bugs pre-release.

Projects

DevMind – Agentic Codebase Intelligence Platform Jan 2026 – Present

Tech Stack: FastAPI, ChromaDB, BM25, cross-encoder, Groq (LLaMA 3.1), LangGraph, tree-sitter, Next.js

- Built an agentic codebase intelligence platform featuring a **hybrid retrieval system** (dense embeddings + BM25 sparse, fused via Reciprocal Rank Fusion, cross-encoder reranked) over tree-sitter AST-parsed GitHub repositories.
- Orchestrated a **LangGraph agent** for multi-hop reasoning: the agent traverses a static call-graph layer across files to resolve cross-file dependencies, returning citation-grounded answers with exact `file:line` references.
- Resolved vector index contamination by debugging asset file pollution; successfully indexed over **500+ files**, **5k+ code chunks**, and **23k+ call-graph edges** across real-world repositories.

DataPulseAI – AI-Powered Backend System Jun 2025 – Oct 2025

Tech Stack: Node.js, Express 5, Firebase Admin SDK, Groq & Gemini APIs, Firestore

- Built a RESTful API for AI-driven data analysis with a **dual-LLM pipeline** (Groq primary, automatic Gemini failover on rate limits/errors) and SSE-streamed chat responses.
- Implemented hybrid authentication combining Firebase Auth with backend-managed sessions (**bcrypt password hashing**, signed httpOnly cookies, single-session enforcement), a Firestore architecture with Firebase Admin SDK as the exclusive writer, and GDPR-compliant soft-delete.

Iron Arc – Cross-Platform Fitness Tracking App Jun 2024 – Feb 2025

Tech Stack: React Native, TypeScript, Node.js, Express, Prisma, Supabase, Firebase, Groq (LLaMA 3.3)

- Built a **cross-platform React Native** fitness app with offline-first architecture using TanStack Query, AsyncStorage persistence, optimistic updates, and background synchronization.
- Engineered an Express + Prisma backend with PostgreSQL (Supabase), Firebase Auth, and a Groq-powered AI trainer that generates validated personalized workout plans.

Skills

Languages: Python, JavaScript/TypeScript

Backend: Node.js, Express.js, FastAPI, REST APIs, JWT, Firebase Admin SDK

Frontend: Next.js, React Native (Expo), React.js, Tailwind CSS

Applied AI: LLM APIs (Gemini & Groq), RAG Pipelines (BM25, ChromaDB, RRF)

Cloud & Tools: Git, GitHub, Gerrit, Linux/Unix, Docker, Jenkins CI/CD, Google Cloud Platform, Postman

Databases: Firebase (Auth/Firestore), MySQL, PostgreSQL, SQLite

Education

Vellore Institute of Technology, Chennai, MTech Integrated Software Engineering

Sept 2021 – Present

CGPA: 8.06/10

Achievements

- Patent (2024):** Smart IoT Backpack with real-time tracking, SOS alerts, and hybrid solar/piezoelectric harvesting.
- Publication (Mar 2026, ICIITCEE):** Co-authored AI grievance redressal system using DistilBERT and EfficientNet; reduced response time 48h→31h and duplicates 18%→6%.
- Google Cloud Arcade (S1–4):** 1.9L+ points and 70+ GCP skill badges.